# DOPE: Dynamic Obstacle Parking Environment for End-to-End Autonomous Parking in CARLA

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***Abstract*— End-to-End (E2E) learning has recently attracted growing interest for autonomous parking, as it enables policies to directly map sensor observations to control commands and potentially learn human-like behaviors in complex environments. In particular, such approaches are expected to generalize better than traditional modular pipelines when trained on sufficiently diverse data. However, existing methods are typically trained and evaluated in static, deterministic environments that fail to capture the variability and interactions present in real-world parking scenarios. Although these limitations have been acknowledged, there is currently no controlled simulation framework for E2E parking that systematically incorporates dynamic and stochastic behaviors for training and evaluation. To address this gap, we introduce Dynamic Obstacle Parking Environment (DOPE), a closed-loop CARLA-based simulation framework with dynamic scenarios designed to study E2E parking under controlled stochasticity and challenging conditions. Rather than inducing failures through adversarial or overly aggressive behaviors, DOPE introduces state-based dynamics that reflect common disturbances encountered in parking environments. Using this framework, we evaluate several state-of-the-art E2E parking models that were not trained under such complex conditions and show that performance degrades noticeably under stochasticity alone and deteriorates further under dynamic interactions with surrounding vehicles. These findings reveal robustness and generalization limitations in current E2E parking models and highlight the need for systematic evaluation under stochastic and interactive parking conditions. The source code is publicly available at `https://github.com/boschresearch/dynamic-obstacle-parking-env/tree/icra2026`.**

## I. Introduction

With one in five reported vehicle crashes occurring in parking lots and garages [1], [2], the development of safer, more reliable autonomous parking systems is a critical priority. To address this, recent work in E2E learning has shown that models can replicate human-like parking maneuvers in simulation, indicating a promising path toward safer and more reliable autonomous parking systems [3], [4], [5], [6].

While promising, the success of these models is almost exclusively demonstrated in simulation, as real-world training and evaluation are costly and potentially unsafe [7]. Although closed-loop simulation is essential for assessing sequential decision-making, current benchmarks for parking are fundamentally limited [3], [4], [5], [6]. They rely on static and deterministic environments where parked vehicles are perfectly aligned and dynamic traffic is either absent or follows simplistic paths. Such conditions can validate geometric accuracy but fail to assess how a policy behaves in interactive scenarios.

This contrasts with real-world parking lots, which are inherently dynamic and unpredictable. Vehicles must navigate narrow aisles, and parked cars are frequently misaligned. These factors demand adaptive decision-making under uncertainty, which cannot be robustly learned or validated in oversimplified settings. Because E2E models are data-driven (trained via Imitation Learning (IL) and/or Reinforcement Learning (RL)), their robustness is shaped by the diversity of the training data and evaluation environments. Consequently, policies that excel in static benchmarks are prone to failing when confronted with stochastic and challenging events, potentially leading to unsafe or unpredictable behaviors [6], [8], [9].

To address this limitation, we introduce DOPE, a dynamic parking scenario suite in CARLA [10] designed to evaluate autonomous parking policies under challenging conditions. To our knowledge, this is among the first frameworks to systematically study the impact of stochastic and interactive dynamics in E2E parking. Rather than inducing failures through adversarial perturbations [11], our objective is to expose the vehicle to complex scenarios commonly encountered in real parking lots, to provide a framework for studying policy robustness under diverse conditions.

DOPE comprises an optional misaligned vehicle setting and three representative classes of parking interaction that are both common and challenging for decision-making: (i) adjacent vehicle departure (a parked vehicle suddenly exits its slot), (ii) on-path following (a trailing vehicle moves within a shared aisle with stochastic stop-and-go behavior), and (iii) transient lane obstruction (a vehicle temporarily blocks access to the target parking slot). These scenarios are designed to capture typical parking lot interactions that require drivers to reason about whether to yield, wait, or proceed under uncertainty, often in conditions of limited visibility and tight spatial constraints. To this end, the proposed scenarios incorporate stochastic and dynamic vehicles that emulate these interactions, as well as variations in parked vehicle alignment that reflect imperfect real-world parking behavior. At the same time, the scenarios remain fully controllable and reproducible to support systematic evaluation. Moreover, DOPE is implemented within a unified framework that enables existing E2E parking models to be integrated with minimal effort.

Using DOPE, we reevaluated representative E2E park-

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ing models and compared their performance against the previous static benchmarks. Our experiments reveal that the introduction of stochasticity alone, such as randomized initializations and misaligned parked vehicles, causes a noticeable degradation in performance. When dynamic vehicles are introduced, performance deteriorates significantly, and collision rates escalate. These findings expose the limitations of existing evaluation practices and underscore the necessity of integrating dynamic and stochastic variability into both training and evaluation pipelines.

The contribution of this paper is as follows:

- We introduce DOPE, a CARLA-based dynamic parking scenario suite that models stochastic and interactive behaviors commonly encountered in real-world parking environments, including misaligned parking layouts and dynamic vehicle interactions.
- We provide a unified and easily extensible evaluation framework that enables seamless integration of existing E2E parking models for closed-loop testing under safety-relevant conditions.
- We conduct a systematic evaluation of representative state-of-the-art E2E parking models under stochastic and dynamic scenarios, revealing substantial performance degradation and failure modes that are not observed in static settings.

## II. Related Works

CARLA [10] is one of the most widely used open-source simulators for autonomous driving research and offers realistic physics, diverse sensor suites, and configurable HD-map-based towns with parking lots. It supports standardized benchmarks, such as leaderboards for route navigation and interactive scenarios, enabling scalable policy training and evaluation under controlled conditions [6].

CARLA Real Traffic Scenarios (CRTS) [12] extends CARLA with a catalog of interactive traffic scenarios, such as highway lane changes and roundabouts, designed for stress testing tactical driving behaviors like merging and right-of-way negotiation. CARLA's built-in Traffic Manager [13] orchestrates background vehicles in autopilot mode to generate dynamic traffic conditions, using rule-based behavior with configurable parameters for traffic flow and rule compliance.

Beyond task-specific benchmarks, there is growing interest in generic scenario generation frameworks for CARLA. ScenarioRunner [14], CARLA's official scenario definition and execution engine, supports scripted traffic scenarios via Python or OpenSCENARIO XML format [15]. Scenic [16] provides a probabilistic domain-specific language for creating and configuring physical objects and agents through abstract, high-level specifications. Complementary to these code-centric approaches, an interactive no-code framework [17] provides a graphical interface for both manual and automated creation of traffic scenarios by representing OpenDRIVE maps [18] as graphs and supporting random sampling of actor types, behaviors, and environmental conditions.

Nevertheless, these tools [12], [13], [14], [16], [17] are primarily designed for structured on-road navigation scenarios and provide limited support for fine-grained maneuvers in unstructured parking environments. Unlike road traffic, parking environments often lack explicit traffic control signals, such as traffic lights and stop signs, and vehicle behavior is less structured and more unpredictable. Existing tools are optimized for structured traffic behaviors such as lane following, junction negotiation, and leading-vehicle interactions, whereas parking scenarios demand finer spatial control and interaction modeling at the level of individual parking slots, including slot occupancy, goal-specific evaluation criteria, reversing maneuvers, and tight spatial interactions within confined spaces. As a result, adapting these frameworks to parking requires substantial additional modeling effort beyond their original design assumptions.

In contrast, [19] studies parking as a multi-agent simulation problem aimed at improving parking efficiency and evaluating parking and unparking trajectories. However, this work is built on the Dragon Lake Parking (DLP) [20] dataset rather than CARLA, and therefore does not provide sensor-level simulation or a CARLA-compatible environment for perception-based evaluation. Moreover, it focuses on planning and trajectory execution from existing vehicle states, rather than providing a full simulation stack for E2E parking policy training and sensor-driven interaction modeling. Similarly, PufferDrive [21] provides a configurable multi-agent driving simulator that can be adapted to parking scenarios, but it does not specifically target sensor-level evaluation required for E2E parking benchmarks.

SEG-Parking [6] advances interactive parking simulation in CARLA by introducing two dynamic scenarios in a narrow aisle. One scenario places a low-priority actor that yields to the Ego Vehicle (EV), which requires the EV to confidently initiate parking. Another scenario features a high-priority actor that proceeds first, forcing the EV to recognize priority and wait patiently. However, the scenarios lack diverse dynamics, such as active perturbations during parking execution.

These limitations motivate the need for dedicated scenario frameworks that explicitly model interactive and stochastic dynamics in parking environments. To this end, we introduce DOPE, a dynamic parking scenario suite based on CARLA that explicitly models complex interactions commonly encountered in parking environments. The framework provides a set of parking scenarios with controlled stochasticity in dynamic vehicles, enabling systematic evaluation of E2E parking policies under interactive conditions. Although currently based on state machines with a single dynamic actor, these scenarios lay the essential groundwork for the automated generation of more general and interactive parking environments.

## III. Dynamic Parking Scenarios

For a fair comparison with existing work and future integration, DOPE reimplements the evaluation environment and interfaces commonly used by prior E2E parking models [3],

[4], [5]. The environment is based on the public parking lot map from `Town04_Opt`, as shown in Figure 1. Candidate target parking slots are located in rows 2 and 3, where odd-numbered slots are reserved for validation and even-numbered slots are used for training. To create a realistic base scene, static actors are spawned in a random selection of non-target slots, occupying at least one-third of the available space. The EV is then initialized at a randomized spawn point near the target, with its heading perpendicular to the parking slots.

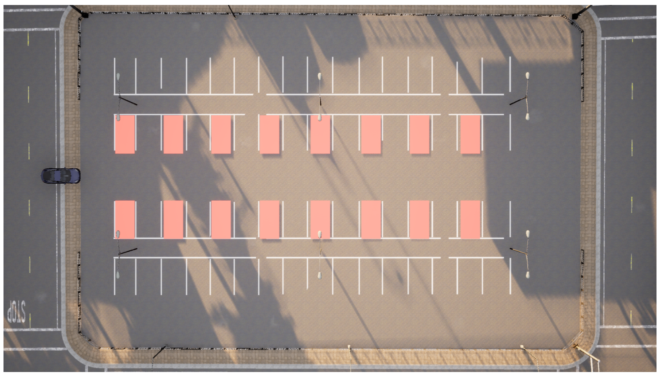


Fig. 1: Layout of `Town04_Opt` and available target parking slots for evaluation

To ensure reproducibility while maintaining controlled stochasticity, we employ a deterministic seeding for all sources of randomness in the simulation. Specifically, a global base seed is defined for each experiment, and individual random components (e.g., EV spawn position, actor placement, and behavior parameters) are generated using the base seed combined with fixed offsets. This design guarantees that each experimental run is fully reproducible while still enabling diverse and systematically varied configurations across trials.

Based on this setup, DOPE introduces a set of configurable scenario modes designed to model safety-relevant interactions in parking environments. The selected scenarios represent common patterns that challenge autonomous decision-making under limited visibility and tight spatial constraints. In particular, they model (i) dynamic occupancy changes caused by departing vehicles (Section III-B), (ii) uncertainty and timing interactions introduced by nearby moving traffic in shared aisles (Section III-C), and (iii) temporary obstruction of access to the target slot (Section III-D). Together with misaligned parked vehicles, these scenarios introduce spatial uncertainty, interaction-driven decision-making, and transient accessibility constraints that are largely absent from conventional static parking benchmarks. These interaction patterns are particularly relevant in parking lots, where narrow aisles, limited visibility, and shared maneuvering spaces frequently require drivers to negotiate right-of-way and adapt to unexpected vehicle behavior.

### A. *Misaligned Vehicles*

To better reflect real-world variability, DOPE includes an optional misaligned vehicles mode in which background vehicles are not perfectly aligned within their parking slots. When enabled, parked actors are initialized with small pose perturbations (i.e., position and orientation offsets), as shown in Figure 2. This option transforms the map from an idealized layout into a more natural parking lot configuration, which may increase the difficulty of parking maneuvers and raise the likelihood of collisions with adjacent vehicles. In addition, the misaligned vehicle setting may reduce the number of parked vehicles due to increased spatial constraints.

It should be noted that these perturbations do not affect the evaluation, as any resulting offset in the final parking position remains within the defined success criteria. Furthermore, the perturbations are bounded so that any overlap does not prevent a successful parking maneuver.

(a) Perfectly parked vehicles (b) Misaligned vehicles

Fig. 2: Comparison between perfectly aligned and misaligned scenarios

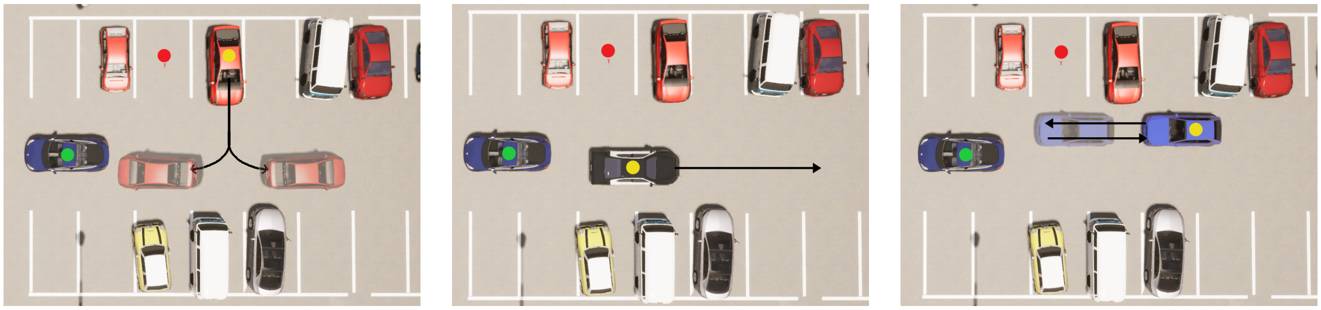
### B. *Adjacent vehicle departure (Drive-Out)*

In the *drive-out* scenario, DOPE models a safety-relevant interaction in which a nearby parked vehicle begins to leave its slot while the EV is approaching or preparing to park. At least one vehicle is spawned adjacent to the target slot, and for each target slot, either the left or right neighboring vehicle (if present) is randomly selected to depart. As illustrated in Figure 3a, the departing actor executes a staged maneuver: it first moves forward out of the slot, then turns toward a randomly sampled direction (i.e., left or right) and stops after completing the turn. This randomness in turn direction may cause the actor to either temporarily block the EV or leave the path unobstructed. This scenario captures a common parking lot hazard in which a vehicle suddenly exits its slot, altering the available free space and requiring the EV policy to react safely and adapt its trajectory online. In particular, the EV should optimally infer whether the departing actor constitutes an active obstruction and decide whether to yield or proceed accordingly.

In the current implementation, all actors are reverse-parked to simplify the drive-out behavior. Future work will extend this scenario to include vehicles that are forward-parked.

### C. *On-path following (Follow)*

In *follow* mode, DOPE introduces a trailing vehicle that shares the driving lane with the EV throughout the parking



(a) Drive-out (b) Follow (c) Block

Fig. 3: Three dynamic actor behaviors in DOPE (The EV is shown in green, the dynamic actor in yellow, and the target parking slot in red)

maneuver. At the start of each run, an actor is spawned either ahead of or behind the EV on the same lateral axis, as illustrated in Figure 3b. The spawn position of the actor is conditioned on the EV's initial position and direction to ensure that it lies along the EV's intended path while avoiding unintended blockage of the EV's maneuver.

Since the evaluated models perform reverse parking, the EV may either directly reverse into the slot or first move forward before reversing, and therefore spawning the actor accordingly maximizes interaction during the maneuver. The actor then moves along the lane using a stochastic stop-and-go policy, alternating between acceleration and braking phases of randomized duration to produce irregular following behavior. This behavior models a common real-world scenario in which a nearby vehicle moves unpredictably within a shared parking-lot lane.

### D. Transient lane obstruction (Block)

In *block* mode, DOPE deploys an actor that intentionally interferes with access to the target slot corridor. At the start of each run, a candidate blocker is spawned near the target lane with a randomized longitudinal offset and an orientation selected to cross the slot entrance. The actor then drives into the corridor and brakes to create a temporary blockage. After a randomized blocking timeout, it transitions to a clearing maneuver: it moves away from the corridor, choosing reverse or forward motion based on the EV's relative position (reverse if the EV is in front, forward if the EV is behind). This creates a transient bottleneck at the slot entrance, requiring the EV to wait, yield, or adapt its approach timing.

In drive-out and block modes, the behavior of dynamic actors is conditioned on their proximity to the EV, incorporating scenario-specific safety-based braking to prevent unrealistic collisions while preserving the intended interaction. In drive-out mode, the departing vehicle brakes when approaching the EV too closely, transitions from the exit phase to the turning phase after moving a short distance, and stops once the turn is completed or exceeds a maximum duration. In block mode, the actor brakes when reaching the slot entrance or approaching the EV, then remains stationary for a randomized dwell interval before clearing the corridor. In follow mode, candidate spawn positions that place the vehicle too close to the EV are rejected, after which the accepted vehicle follows a stochastic stop-and-go policy. These safety constraints encourage the EV to adapt to the current situation rather than inducing failures through unavoidable collisions, although they may occasionally limit the intended dynamic obstruction. For example, in drive-out mode, the departing actor may remain stationary if the EV blocks its exit path.

## IV. Benchmarking

To quantify the impact of dynamic interactions, we benchmark three publicly available, imitation learning-based models: E2E Parking [3], CAA Policy [4], and DDP [5]. We used the official checkpoints released by the original authors (4-epoch versions for all models) to ensure a fair and reproducible comparison. All experiments were conducted in CARLA 0.9.11, consistent with prior work. The evaluation protocol consists of 6 trials for each of the 16 target parking slots, varying the EV's initial spawn position and orientation for each trial.

It is important to note that the publicly available checkpoint for E2E Parking exhibits lower performance than the results originally reported by its authors. Therefore, its results should be interpreted as a reference. For Dino-Diffusion Parking (DDP), we used the DDP-emb+seg configuration. The RL-based SEG Parking model [6] was excluded as its implementation is not publicly available.

We evaluate the models across a range of increasingly complex settings, presented in Table I:

- Reported: The original performance metrics as published in the source papers.
- Reproduced: Our results from replicating the original experiments. These use the authors' pseudo-deterministic setup, where the EV spawn position and heading are sampled within a predefined range.
- Static Baseline: To isolate the effect of initial state stochasticity, this setting modifies only one variable from the reproduced setup. The environment remains static (no misaligned or dynamic vehicles), but we replace the pseudo-deterministic spawning with a fully

randomized position and heading within the same predefined range used in prior work.
- DOPE Scenarios: We then evaluate the models (without retraining) in four dynamic DOPE settings: (i) Misaligned vehicles only, and (ii) Misaligned + the three dynamic scenarios (Drive-Out, Follow, and Block). Misaligned vehicles are included in all dynamic tests to reflect a realistic spatial layout.

Model performance is measured using four standard metrics from prior work:

- Target Success Rate (TSR) measures the fraction of trials where the EV parks successfully in the correct slot. A trial is considered successful if the distance between the center of the EV and the center of the target slot is within 0.6 m laterally and 1.0 m longitudinally, and the orientation error is less than $10^\circ$.
- Non-target Success Rate (NTSR) measures the fraction of trials where the EV parks in the wrong slot.
- Collision Rate (CR) measures the fraction of trials with any collision.
- Timeout Rate (TR) measures the fraction of trials where the maneuver is not completed within 900 simulation frames.

As shown in Table I, we observe a consistent performance gap between the results reported in prior work, our reproduced results, and the proposed static baseline. While the reproduced results generally follow the trends reported in the original papers under pseudo-deterministic settings, a noticeable performance drop is observed when moving to our baseline setting. Specifically, all models exhibit lower TSR and higher CR under the baseline compared to both reported and reproduced results.

This degradation indicates that even mild stochasticity introduced through randomized ego initialization leads to a distribution shift from the training conditions, resulting in reduced performance. The gap between reproduced and baseline results suggests that existing E2E parking models have limited generalization beyond pseudo-deterministic setups, even when the environment remains static.

Comparing the baseline with the misaligned setting, we observe a further decline in performance, characterized by decreased TSR and increased CR. This highlights the impact of spatial perturbations in parked vehicles, where misalignment introduces tighter constraints and increases the likelihood of collision.

The introduction of dynamic vehicles leads to a significant additional drop in performance across all models. As interaction complexity increases, TSR decreases sharply while CR rises, indicating that current models struggle to handle dynamic and interactive environments. In particular, scenarios with stronger obstruction, such as transient blocking of the target slot, result in the most severe degradation. For example, TSR drops to 9.64% for E2E Parking, 19.27% for CAA Policy, and 24.48% for DDP.

The observed performance degradation is consistent with well-known failure modes in IL, such as covariate shift [8], in which performance deteriorates when a policy encounters states that are out of distribution relative to its training data. By presenting the policy with challenges it was likely not trained to handle, these scenarios create conditions under which compounding errors are expected to arise. While a rigorous analysis of this mismatch is beyond the scope of this paper, our results demonstrate that current models are highly vulnerable to such generalization failures.

Among the evaluated models, DDP consistently achieves the highest performance under DOPE, maintaining relatively higher TSR and exhibiting zero NTSR across all scenarios, indicating more precise target selection. However, even DDP experiences substantial degradation under dynamic conditions, underscoring the overall challenge of robust E2E parking in interactive environments.

These results reveal that current E2E parking models rely heavily on deterministic assumptions and struggle to generalize beyond the conditions present in conventional evaluation settings. This highlights the need for incorporating dynamic and stochastic scenarios during both training and evaluation.

## V. Conclusion

In this work, we present DOPE, a closed-loop simulation environment with dynamic scenarios built on CARLA for evaluating E2E autonomous parking models under interactive conditions. Unlike existing benchmarks that rely on static and deterministic setups, DOPE introduces controlled stochasticity and dynamic interactions that better reflect real-world parking environments while maintaining reproducibility. By evaluating state-of-the-art E2E parking models, we demonstrate that performance degrades significantly when transitioning from static to stochastic and dynamic scenarios, highlighting the limitations of current approaches and the importance of evaluating parking policies under diverse stochastic and interactive conditions.

DOPE provides a practical step toward bridging the gap between simplified simulation environments and real-world parking complexity. As an easily integrable simulation framework with diverse interactive conditions, it offers a useful tool for developing more robust and generalizable parking policies.

In future work, we plan to extend DOPE along several directions. First, we aim to support a wider range of maps to increase environmental diversity. Second, we plan to design additional dynamic scenarios that further challenge E2E parking models under more complex interactions that incorporate multiple dynamic vehicles. Finally, we will explore agent-based control for actors, replacing the current state-machine-driven behaviors to enable more realistic and adaptive interactions.

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TABLE I: Performance comparison of E2E parking models under different scenarios.

| Model | Setting | TSR (%) ↑ | NTSR (%) ↓ | CR (%) ↓ | TR (%) ↓ |
|---|---|---|---|---|---|
| E2E Parking [3] | Reported[a] | 91.41 | 3.39 | 2.08 | 1.04 |
| | Reproduced[b] | 82.55 ± 4.16 | 4.17 ± 1.94 | 7.29 ± 2.65 | 2.86 ± 1.36 |
| | Static Baseline[c] | 72.14 ± 4.23 | 3.65 ± 1.74 | 14.58 ± 3.46 | 2.86 ± 1.22 |
| | Misaligned | 61.46 ± 4.42 | 4.17 ± 1.26 | 26.30 ± 4.01 | 3.65 ± 1.54 |
| | Misaligned + Drive-Out | 16.67 ± 3.42 | 4.43 ± 0.77 | 71.88 ± 4.59 | 5.73 ± 1.81 |
| | Misaligned + Follow | 10.42 ± 1.93 | 3.12 ± 0.74 | 78.12 ± 3.54 | 6.51 ± 2.01 |
| | Misaligned + Block | 9.64 ± 3.02 | 5.21 ± 1.92 | 79.95 ± 4.50 | 4.17 ± 2.07 |
| CAA Policy [4] | Reported[a] | 87.5 ± 3.7 | 0.8 ± 0.6 | 3.5 ± 2.6 | 3.8 ± 2.2 |
| | Reproduced[b] | 87.24 ± 4.12 | 0.00 ± 0.00 | 5.21 ± 2.00 | 4.95 ± 2.55 |
| | Static Baseline[c] | 71.35 ± 3.56 | 1.56 ± 0.37 | 22.92 ± 2.26 | 2.86 ± 1.51 |
| | Misaligned | 68.75 ± 3.53 | 1.82 ± 0.93 | 22.14 ± 2.55 | 3.91 ± 2.19 |
| | Misaligned + Drive-Out | 43.23 ± 3.52 | 0.26 ± 0.18 | 54.17 ± 3.03 | 1.30 ± 0.92 |
| | Misaligned + Follow | 28.65 ± 3.00 | 1.56 ± 0.37 | 67.45 ± 2.72 | 0.78 ± 0.55 |
| | Misaligned + Block | 19.27 ± 1.47 | 0.78 ± 0.18 | 76.56 ± 1.32 | 1.30 ± 0.77 |
| DDP [5][d] | Reported[a] | 92 ± 11 | 0 ± 0 | 0 ± 0 | 4 ± 10 |
| | Reproduced[b] | 92.71 ± 1.32 | 0.00 ± 0.00 | 0.00 ± 0.00 | 3.91 ± 0.92 |
| | Static Baseline[c] | 84.38 ± 1.27 | 0.00 ± 0.00 | 6.25 ± 0.00 | 5.99 ± 1.43 |
| | Misaligned | 80.47 ± 1.13 | 0.00 ± 0.00 | 6.77 ± 0.21 | 6.25 ± 1.27 |
| | Misaligned + Drive-Out | 57.03 ± 1.03 | 0.00 ± 0.00 | 38.02 ± 0.21 | 2.08 ± 0.58 |
| | Misaligned + Follow | 41.15 ± 1.07 | 0.00 ± 0.00 | 56.25 ± 0.37 | 0.26 ± 0.18 |
| | Misaligned + Block | 24.48 ± 0.95 | 0.00 ± 0.00 | 72.66 ± 0.77 | 0.78 ± 0.18 |

[a] Reported results correspond to the original evaluation reported in prior work, which use pseudo-deterministic ego initialization and static environments.
[b] Reproduced results are obtained using the publicly available checkpoints under the same pseudo-deterministic setup for fair comparison.
[c] Static baseline refers to a static environment without dynamic interactions or misaligned vehicles, where the ego vehicle is initialized at randomly sampled positions and orientations near the target slot.
[d] DDP-emb+seg
TSR: Target Success Rate; NTSR: Non-Target Success Rate; CR: Collision Rate; TR: Timeout Rate.
↑ indicates higher is better; ↓ indicates lower is better.


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